Task 2 - Automatic Street Light System Using LDR

Code:

```
int sensorPin = A0;
int sensorValue = 0;
int led = 9;

void setup()
 {
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);
  if(sensorValue < 150){
    Serial.println("Time to shine, it's dark!");
    digitalWrite(led,HIGH);
  }
  else
{
    Serial.println("No need for light, it's bright out!");
    digitalWrite(led, LOW);
  }
  delay(1000);
}
```

Explanation of the Code

1. Variable Declaration

int sensorPin = A0;

- The LDR sensor is connected to analog pin A0 of Arduino.
- This pin reads the analog voltage from the sensor.

int sensorValue = 0;

- This variable stores the value read from the LDR sensor.

int led = 9;

- The LED is connected to digital pin 9.

2. setup() Function

void setup() {

- The setup() function runs only once when the Arduino starts.

pinMode(led, OUTPUT);

- Sets pin 9 as an OUTPUT pin because the LED needs output signals.

Serial.begin(9600);

- Starts serial communication at 9600 baud rate.
- Used to display sensor values and messages on the Serial Monitor.

3. loop() Function

void loop(){

- The loop() function runs continuously again and again.

4. Reading Sensor Value

sensorValue = analogRead(sensorPin);

- Reads the analog value from the LDR sensor.
- The value ranges from 0 to 1023.

Working of LDR

- In bright light → resistance decreases → higher analog value.

- In darkness → resistance increases → lower analog value.

5. Displaying Sensor Value

Serial.println(sensorValue);

- Prints the sensor value on the Serial Monitor for monitoring and testing.

6. Condition for Darkness

if(sensorValue < 150)

- Checks whether the light intensity is low.
- If the sensor value is less than 150, the environment is considered dark.

7. LED ON Condition

digitalWrite(led,HIGH);

- Turns the LED ON when darkness is detected.

Serial.println("Time to shine, it's dark!");

- Displays a message on the Serial Monitor.

8. LED OFF Condition

digitalWrite(led, LOW);

- Turns the LED OFF when the environment is bright.

Serial.println("No need for light, it's bright out!");

- Displays a message indicating sufficient light.

9. Delay Function

delay(1000);

- Waits for 1 second before repeating the loop.
- Prevents continuous rapid printing on the Serial Monitor.